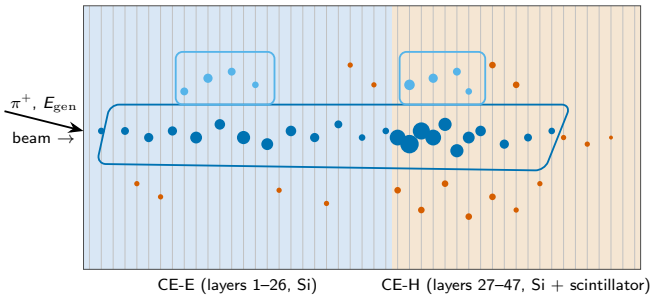
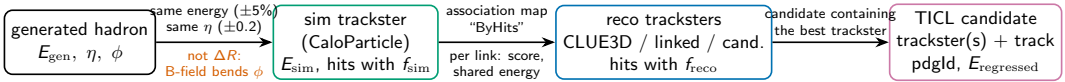
HARP: what we look at in HGCal



- **layer clusters** (dots): CLUE clusters of rec-hits on one layer; size $\propto$ energy
- **best trackster**: the CLUE3D trackster best matched to the hadron
- **other tracksters**: further tracksters sharing the hadron's hits (fragments)
- **no trackster**: layer clusters that CLUE3D did not attach to any trackster
- everything is measured versus E_{gen}, per hadron type ($\pi^0, \pi^{\pm}, K^{\pm}, p, \bar{p}$)

The matching chain: gen → sim → reco → candidate



E_{sim} = regressed energy of the sim trackster
= the CaloParticle energy = E_{gen}

best = lowest score
linked = score < cut, shared $E \geq$ min
all = every trackster of the event

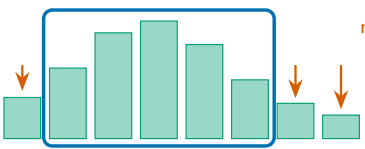
The `HadronTruthMatcher` fills one `HadronMatch` per generated hadron with all of these; every analyzer reads it.

The association score: hit bookkeeping, not a probability

$$\mathrm{score}(S \to R) = \frac{\sum_{\mathrm{hits}} (f_{\mathrm{sim}} - f_{\mathrm{reco}})^2 E_{\mathrm{hit}}^2}{\sum_{\mathrm{hits}} f_{\mathrm{sim}}^2 E_{\mathrm{hit}}^2}$$

hits of the sim trackster S ($f_{\mathrm{sim}} = 1$)

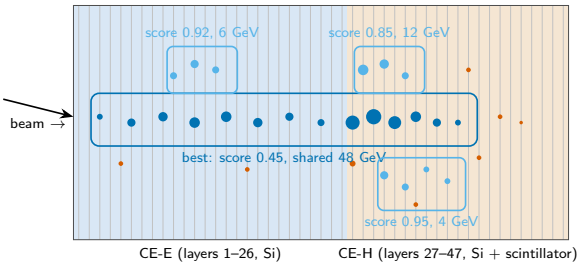
reco trackster R contains hits 2–6



- f: fraction of the hit's energy assigned to the object
- hit in S but not in R: penalty E_{hit}^2; hit in both: no penalty
- bounded: $0 = R$ contains all of S, $1 =$ nothing shared
- for all-or-nothing hits: **score** $\approx 1-$ **(E^2-weighted fraction of S captured by R)**
- E^2 weighting: dominated by the few highest-energy hits
- reco→sim score: same formula normalised to R's hits

Computed in CMSSW from the simulated hit fractions (`AllTracksterToSimTracksterAssociatorsByHitsProducer`); deterministic, no training.

Best, linked, all: the reco energy definitions

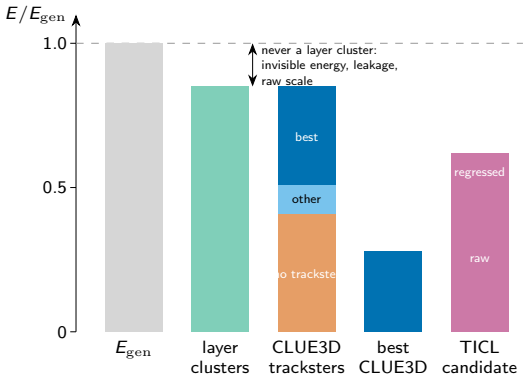


Response $R = E_{\mathrm{reco}}/E_{\mathrm{gen}}$ with E_{reco} one of:

- **bestRaw**: raw energy of the lowest-score trackster
- **linkedRaw**: sum over all tracksters sharing hits (score $<$ `linkScoreCut`, default 1)
- **allRaw**: every trackster of the event (= linked for a gun, differs with pile-up)
- **simRaw**: the sim trackster, i.e. what the hadron deposited
- **layerClusters**: all layer clusters, clustered or not
- **candidate**: TICL candidate (regressed); **candidateRaw**: its raw sum

"raw" = sum of layer-cluster energies, no calibration. The plots show scale $\langle R \rangle$ and resolution $\sigma(R)/\langle R \rangle$ vs E_{gen} from Gaussian fits in each E_{gen} bin.

Where the energy goes: the energy budget

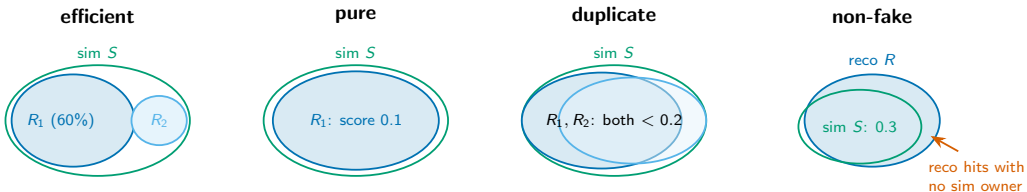


`EnergyBudgetAnalyzer` classifies every layer cluster of the event:

- in the best trackster, in another trackster, or in no trackster (shared clusters split $1/n$);
- as fractions of the layer-cluster energy (stack to 1) or divided by E_{gen} (stack to the layer-cluster response);
- split by compartment (CE-E / CE-H), per layer in E_{gen} windows, and versus the layer-cluster energy.

Example π^+, 85 GeV: 0.85 reaches layer clusters, 0.51 is in tracksters, 0.28 in the best one; linking brings the candidate trackster to 0.46 raw, the regression to 0.62.

Efficiency, purity, duplicate, fake (CMSSW HGCalValidation definitions)



$\geq$ 1 trackster with shared $E \geq 0.5\,E_{\mathrm{sim}}$

$\geq$ 1 trackster with score < 0.2

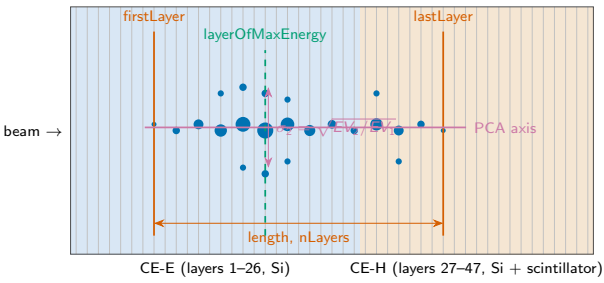
$\geq$ 2 tracksters with score < 0.2

reco trackster with reco$\rightarrow$sim score $<$ 0.6

quantity	per	CMSSW parameter (default)	HARP plot
efficiency	sim trackster	`minTSTSharedEneFracEfficiency` (0.5)	`matching/efficient/vsE`
purity, duplicate	sim trackster	`maxSimToRecoScoreForPurity/Duplicate` (0.2)	`matching/pure, duplicate/vsE`
non-fake	reco trackster	`maxRecoToSimScoreForNonFake` (0.6)	`matching/nonFake/vsE, vsRawE`

Purity asks for *one* trackster containing $\sim$ 80% of the hadron's E^2-weighted hits: a hadron shower split into several good tracksters fails it. HARP adds `hasReco` (any link at all) as the lower bar.

Shower variables for identification (per trackster)



From `ShowerVariablesProducer` (`pid/` plots, best trackster):

- depth: **firstLayer**, **lastLayer**, **length** (cm), **nLayers**, **layerOfMaxEnergy**, **rmsLayer**
- compartments: **fractionEE** = $E_{\text{CE-E}}/(E_{\text{CE-E}} + E_{\text{CE-H}})$, **emFraction** = `raw_em_energy/raw_energy`
- shape: **sigma2**, **sigma3** from the PCA eigenvalues, **maxLayerEnergyFraction**, **nLayerClusters**
- plus the TICL **candidate pdgId** versus E_{gen} (the identification TICL already makes)

π^0: compact, all in CE-E, $\sim$25 layers. Charged hadrons: start anywhere, reach CE-H, wide spread of nLayers.